

Industrial Internship Report on Banking Information System

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Executive Summary

This report presents the work completed during the Industrial Internship organized by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT). The internship duration was six weeks and focused on solving an industry-oriented problem using software development concepts.

The project undertaken during this internship was "**Banking Information System**". The objective of the project was to develop a banking management application that allows users to create accounts, deposit money, withdraw money, and view account details efficiently.

The project was developed using Java programming language and Object-Oriented Programming concepts. The system provides a simple and secure way to manage customer banking information.

This internship provided practical exposure to software development methodologies, programming logic, debugging, testing, and project documentation. It helped improve my technical and problem-solving skills and gave me valuable experience in developing real-world applications.

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1. Preface

The six-week internship program provided me with an opportunity to work on a practical software development project. The project assigned was the development of a Banking Information System.

The main purpose of this project was to design a user-friendly application capable of maintaining customer account information and performing basic banking operations such as deposits, withdrawals, and balance inquiries.

This internship helped me understand the importance of software design, data management, testing, and documentation in real-world projects. I gained valuable knowledge of Java programming, Object-Oriented Programming concepts, and application development.

I sincerely thank upskill Campus, The IoT Academy, UniConverge Technologies Pvt Ltd, mentors, and trainers for providing guidance and support throughout the internship.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is a technology company established in 2013 that provides digital transformation solutions using technologies such as IoT, Cloud Computing, Machine Learning, Cyber Security, Java Full Stack Development, Python, and Front-End Development.

2.2 About upskill Campus

upskill Campus is a career development platform that helps students gain industry-relevant skills through internships, certifications, projects, and professional training programs.

2.3 About The IoT Academy

The IoT Academy is the EdTech division of UCT and offers professional certification programs in collaboration with prestigious institutions including IIT Kanpur, IIT Roorkee, and IIT Guwahati.

2.4 Objectives of this Internship Program

- To gain practical industry experience.
- To understand software development processes.
- To improve programming and problem-solving skills.
- To work on a real-world project.
- To enhance career readiness and technical expertise.

2.5 References

- [1] Java Documentation
- [2] Oracle Java Tutorials
- [3] GeeksforGeeks Java Programming Tutorials
- [4] Object-Oriented Programming Concepts
- [5] Banking Management System Case Studies

2.6 Glossary

Term	Acronym
Object-Oriented Programming	OOP
User Interface	UI
Integrated Development Environment	IDE
Java Development Kit	JDK
Banking Information System	BIS

3. Problem Statement

Traditional banking record management can be time-consuming and prone to errors when handled manually. Managing customer accounts, deposits, withdrawals, and account information requires an efficient and reliable system.

The objective of this project is to develop a Banking Information System that allows users to:

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Create bank accounts.

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-

Store customer information.

-
-

Deposit money.

-
-

Withdraw money.

-
-

Check account details.

-

-

Maintain accurate account balances.

-

The system should provide a simple, secure, and efficient solution for managing banking operations.

4. Existing and Proposed Solution

Existing Solutions

Traditional banking systems often rely on complex enterprise-level software that may be expensive and difficult to implement for small organizations or educational purposes.

Limitations

-

High implementation cost.

-
-

Complex architecture.

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-

Requires extensive infrastructure.

-
-

Difficult for beginners to understand.

-

Proposed Solution

A Java-based Banking Information System that provides basic banking functionalities through a simple console-based application.

Features

-

Account Creation

-
-

Deposit Money

-
-

Withdraw Money

-
-

View Account Information

-
-

Secure Balance Management

-

Advantages

-

Easy to use.

-
-

Cost-effective.

-
-

Fast transaction processing.

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-

Simple implementation using Java.

-

4.1 Code Submission (GitHub Link)

GitHub Repository:

(Paste your GitHub repository link here)

4.2 Report Submission (GitHub Link)

(Paste report repository link here)

5. Proposed Design / Model

The Banking Information System follows a structured workflow to manage customer accounts and banking transactions.

5.1 High Level Diagram

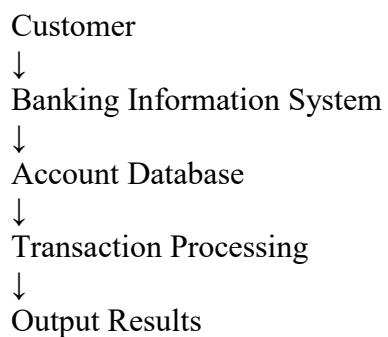
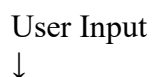


Figure 1: High Level Diagram of Banking Information System

5.2 Low Level Diagram



Account Creation Module
↓
Account Validation
↓
Deposit / Withdrawal Module
↓
Balance Update
↓
Display Results

Figure 2: Low Level Diagram of Banking Information System

5.3 Interfaces

Main Menu

1.

Create Account

2.

3.

Deposit Money

4.

5.

Withdraw Money

6.

7.

View Account Details

8.

9.

Exit

10.

Input Parameters

-

Account Holder Name

-

-

Account Number

-

-

Initial Balance

-

-

Transaction Amount

-

Output Parameters

-

Updated Balance

-

-

Transaction Status

-

-

Account Information

-

6. Performance Test

6.1 Test Plan / Test Cases

Test Case	Expected Result
Create Account	Account created successfully
Deposit Money	Balance updated
Withdraw Available Amount	Transaction successful
Withdraw Excess Amount	Insufficient balance message
View Account	Account details displayed

6.2 Test Procedure

1.

Start the application.

2.

3.

Create a customer account.

4.

5.

Perform deposit transactions.

6.

7.

Perform withdrawal transactions.

8.

9.

Verify balance updates.

10.

11.

Display account details.

12.

13.

Validate output results.

14.

6.3 Performance Outcome

-

Successful account creation.

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-

Accurate balance calculations.

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-

Fast transaction execution.

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-

Proper validation for insufficient balance.

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Reliable account information retrieval.

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The system performed efficiently under all test conditions and produced accurate results.

7. My Learnings

During this internship, I learned:

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Java Programming Fundamentals.

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-

Object-Oriented Programming Concepts.

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Class and Object Implementation.

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Data Management Techniques.

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Software Testing and Debugging.

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-

GitHub Repository Management.

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-

Technical Documentation.

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-

Problem-Solving Skills.

-

This project improved my understanding of software development and enhanced my confidence in developing real-world applications.

8. Future Work Scope

The current Banking Information System can be enhanced by implementing the following features:

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Graphical User Interface (GUI).

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-

Database Integration using MySQL.

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User Authentication and Security.

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Transaction History Management.

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Online Banking Features.

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Fund Transfer Functionality.

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Mobile Application Support.

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-

Report Generation System.

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These enhancements can transform the project into a fully functional banking management solution suitable for real-world applications.



Conclusion

The Banking Information System was successfully designed, developed, and tested during the internship period. The project achieved its objectives by providing an efficient solution for managing customer accounts and banking transactions. The internship provided valuable practical experience and strengthened my programming, analytical, and software development skills.